4. FUNCTIONAL REQUIREMENTS

This section defines the functional requirement baseline for the VMU power-split mode logic. The system requirements preserve the supervisory intent at vehicle-function level, the high-level software requirements organize the software behavior by state family, and the low-level software requirements capture each executable transition individually.

4.1. High-level system functional requirements

4.1.1 SysHLR01 - Power-Split Supervisory Mode Selection

[SysHLR01] Power-Split Supervisory Mode Selection	
Requirement statement (shall)	The VMU shall select exactly one operating mode from {standstill mode, EV mode, regenerative-braking mode, Start mode, ICE mode, and Hybrid mode} at each mode-logic execution cycle by evaluating vehicle speed [km/h], requested power [kW], battery state of charge [-], and engine speed [rpm].
Conditions / Applicability	Active whenever Control/Mode Logic is executed in nominal mode-management operation.
Stakeholder Priority	High
Rationale	Establishes the single-state supervisory decision expected by the model and the software.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.1.2 SysHLR02 - Deterministic Powertrain Enable Coordination

[SysHLR02] Deterministic Powertrain Enable Coordination	
Requirement statement (shall)	For each selected operating mode, the VMU shall command deterministic motor, generator, and engine enable requests according to the mode-dependent pattern defined by the model: standstill {0,0,0}, EV {1,0,0}, regenerative braking {1,0,0}, Start {1,1,0}, ICE {0,1,1}, and Hybrid {1,1,1}.
Conditions / Applicability	Applies in each healthy mode-logic cycle after the next mode has been resolved.
Stakeholder Priority	High
Rationale	Ensures supervisory mode selection and plant-facing enable outputs remain consistent.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.1.3 SysHLR03 - Calibratable and Hysteretic Transition Strategy

[SysHLR03] Calibratable and Hysteretic Transition Strategy	
Requirement statement (shall)	The VMU shall evaluate mode transitions using the named calibrations ENG_ON, ENG_OFF, SPEED_STOP, SPEED_REGEN, SPEED_EV_MAX, PDEM_REGEN, PDEM_STOP_LOW, PDEM_STOP_HIGH, PDEM_HYB_IN,

	PDEM_HYB_OUT, PDEM_HYB_MID, PDEM_HYB_LOW, SOC_EV_IN, SOC_EV_OUT, SOC_MID, and SOC_LOW while preserving the modeled hysteresis on battery state of charge and engine speed thresholds.
Conditions / Applicability	Applies to every transition decision in the mode logic.
Stakeholder Priority	High
Rationale	Keeps the requirement baseline aligned with the calibrated model and preserves the intended hysteresis behavior.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.2. High-level software functional requirements

The high-level software requirements below organize the software representation by initialization, output mapping, state-family decision logic, and software-defined behaviors that remain outside the modeled transition set. In this section, the software modes correspond to standstill, EV, regenerative braking, start, ICE, and hybrid behavior represented by the model. These requirements act as parents for the transition-level low-level software requirements defined in Section 4.3.

4.2.1 SwHLR01 - Initialization Behavior

[SwHLR01] Initialization Behavior	
Requirement statement (shall)	The software shall initialize the active software mode to the standstill mode before the first normal mode-evaluation cycle.
Conditions / Applicability	Applies during software initialization and before the first periodic mode-evaluation cycle.
Derived from	SysHLR01
Stakeholder Priority	High
Rationale	Creates a deterministic starting point for the software representation of the state machine.
Related UC(s)	UC1, UC6

4.2.2 SwHLR02 - Deterministic Output Mapping by Active Mode

[SwHLR02] Deterministic Output Mapping by Active Mode	
Requirement statement (shall)	The software shall command the enable outputs associated with the resolved active software mode according to the mode-to-output mapping of the source model.
Conditions / Applicability	Applies after each mode-resolution decision.
Derived from	SysHLR02
Stakeholder Priority	High
Rationale	Separates output assignment from transition selection and avoids duplicating output obligations in every low-level transition requirement.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.2.3 SwHLR03 - Standstill Transition Family

[SwHLR03] Standstill Mode Transition Family	
Requirement statement (shall)	While the active software mode is the standstill mode, the software shall evaluate exits to EV mode and Start mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode represents standstill behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the standstill decision family once and allocates the detailed guards to derived low-level requirements.
Related UC(s)	UC1

4.2.4 SwHLR04 - EV Transition Family

[SwHLR04] EV Mode Transition Family	
Requirement statement (shall)	While the active software mode is EV mode, the software shall evaluate exits to regenerative-braking mode, Start mode, and standstill mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode represents EV behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the EV decision family once and allocates the detailed guards to derived low-level requirements.
Related UC(s)	UC2, UC3, UC4, UC6

4.2.5 SwHLR05 - Regenerative-Braking Transition Family

[SwHLR05] Regenerative-Braking Transition Family	
Requirement statement (shall)	While the active software mode is the regenerative-braking mode, the software shall evaluate exits to Start mode, standstill mode, and EV mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode represents regenerative-braking behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the regenerative-braking decision family once and allocates the detailed guards to derived low-level requirements.
Related UC(s)	UC2, UC3, UC4, UC6

4.2.6 SwHLR06 - Engine-Supported External Exit Family

[SwHLR06] Engine-Supported External Exit Family	
Requirement statement (shall)	While the active software mode is Start, ICE, or Hybrid, the software shall evaluate exits to regenerative-braking mode, EV mode, and standstill mode before any internal transition within the engine-supported mode representation.

Conditions / Applicability	Applies while the active software mode represents Start, ICE, or Hybrid behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the common external-exit policy of Motion_mode_ICE and provides the parent requirement for the nine derived low-level exits.
Related UC(s)	UC2, UC4, UC6

4.2.7 SwHLR07 - Start Mode Internal Transition Family

[SwHLR07] Start Mode Internal Transition Family	
Requirement statement (shall)	While the active software mode is Start and no higher-priority engine-supported external exit is active, the software shall evaluate internal transitions to Hybrid mode and ICE mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode is Start and the external-exit family does not trigger.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the internal Start_mode decision family and provides the parent requirement for the derived low-level transitions.
Related UC(s)	UC3, UC5

4.2.8 SwHLR08 - ICE Mode Internal Transition Family

[SwHLR08] ICE Mode Internal Transition Family	
Requirement statement (shall)	While the active software mode is ICE and no higher-priority engine-supported external exit is active, the software shall evaluate internal transitions to Start mode and Hybrid mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode is ICE and the external-exit family does not trigger.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the internal ICE_mode decision family and provides the parent requirement for the derived low-level transitions.
Related UC(s)	UC3, UC5

4.2.9 SwHLR09 - Hybrid Mode Internal Transition Family

[SwHLR09] Hybrid Mode Internal Transition Family	
Requirement statement (shall)	While the active software mode is Hybrid and no higher-priority engine-supported external exit is active, the software shall evaluate internal transitions to Start mode and ICE mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode is Hybrid and the external-exit family does not trigger.

Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the internal Hybrid_mode decision family and provides the parent requirement for the derived low-level transitions.
Related UC(s)	UC3, UC5

4.2.10 SwHLR10 - Initialization and Invalid-Mode Recovery Outside the Modeled Transition Set

[SwHLR10] Initialization and Invalid-Mode Recovery Outside the Modeled Transition Set	
Requirement statement (shall)	The software shall treat initialization to the standstill mode and recovery from an invalid active-mode value as software-defined behaviors outside the 23 modeled transitions.
Conditions / Applicability	Applies during initialization and whenever the stored active mode does not match a valid software mode.
Derived from	SysHLR01
Stakeholder Priority	High
Rationale	Separates startup and defensive recovery behavior from the modeled transition set while keeping the implementation behavior explicit in the requirement baseline.
Related UC(s)	UC1, UC6

4.3. Low-level software functional requirements

The low-level software requirements below provide one requirement per executable transition. The three chart-level exits from Motion_mode_ICE are allocated in software as one requirement per active software mode in the start, ICE, and hybrid branches, so the low-level section contains 23 transition requirements. This count excludes initialization to the standstill mode and recovery from an invalid active-mode value, because those behaviors are software-defined and are not modeled transitions.

4.3.1 SwLLR01 - Standstill Mode to EV Mode

[SwLLR01] Standstill Mode to EV Mode	
Requirement statement (shall)	When the active software mode is the standstill mode and speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is the standstill mode. This is the highest-priority matching exit.
Derived from	SwHLR02, SwHLR03
Stakeholder Priority	High
Rationale	Implements the admissible EV launch path from standstill.
Related UC(s)	UC1, UC2

4.3.2 SwLLR02 - Standstill Mode to Start Mode

[SwLLR02] Standstill Mode to Start Mode

Requirement statement (shall)	When the active software mode is the standstill mode and speed > SPEED_STOP && (speed > SPEED_EV_MAX || SOC < SOC_EV_IN), the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is the standstill mode, after the higher-priority EV exit has been excluded. Entry into Motion_mode_ICE is represented in software by Start mode.
Derived from	SwHLR02, SwHLR03
Stakeholder Priority	High
Rationale	Implements the direct launch path into engine-supported propulsion when the EV window is not valid.
Related UC(s)	UC1, UC3

4.3.3 SwLLR03 - EV Mode to Regenerative-Braking Mode

[SwLLR03] EV Mode to Regenerative-Braking Mode	
Requirement statement (shall)	When the active software mode is EV mode and speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is EV mode. This is the highest-priority EV exit.
Derived from	SwHLR02, SwHLR04
Stakeholder Priority	High
Rationale	Implements entry into regenerative braking from EV operation.
Related UC(s)	UC4

4.3.4 SwLLR04 - EV Mode to Start Mode

[SwLLR04] EV Mode to Start Mode	
Requirement statement (shall)	When the active software mode is EV mode and (speed > SPEED_EV_MAX || P_dem >= PDEM_HYB_IN || SOC < SOC_EV_OUT), the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is EV mode, after the higher-priority regenerative-braking exit has been excluded. Entry into Motion_mode_ICE is represented in software by Start mode.
Derived from	SwHLR02, SwHLR04
Stakeholder Priority	High
Rationale	Implements the EV exit to engine-supported propulsion when the EV window closes or demand increases.
Related UC(s)	UC3

4.3.5 SwLLR05 - EV Mode to Standstill Mode

[SwLLR05] EV Mode to Standstill Mode	
Requirement statement (shall)	When the active software mode is EV mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.

Conditions / Applicability	Applies while the active software mode is EV mode, after the higher-priority regenerative-braking and engine-supported exits have been excluded.
Derived from	SwHLR02, SwHLR04
Stakeholder Priority	High
Rationale	Implements the EV return to standstill in the neutral stop-demand window.
Related UC(s)	UC6

4.3.6 SwLLR06 - Regenerative-Braking Mode to Start Mode

[SwLLR06] Regenerative-Braking Mode to Start Mode	
Requirement statement (shall)	When the active software mode is the regenerative-braking mode and (((speed > SPEED_EV_MAX) && (P_dem >= PDEM_STOP_LOW)) || SOC < SOC_EV_OUT), the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is the regenerative-braking mode. This is the highest-priority regenerative-braking exit. Entry into Motion_mode_ICE is represented in software by Start mode.
Derived from	SwHLR02, SwHLR05
Stakeholder Priority	High
Rationale	Implements the return from regenerative braking to engine-supported propulsion when the EV window is no longer valid.
Related UC(s)	UC3, UC4

4.3.7 SwLLR07 - Regenerative-Braking Mode to Standstill Mode

[SwLLR07] Regenerative-Braking Mode to Standstill Mode	
Requirement statement (shall)	When the active software mode is the regenerative-braking mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is the regenerative-braking mode, after the higher-priority engine-supported exit has been excluded.
Derived from	SwHLR02, SwHLR05
Stakeholder Priority	High
Rationale	Implements braking completion into standstill from the regenerative-braking mode.
Related UC(s)	UC6

4.3.8 SwLLR08 - Regenerative-Braking Mode to EV Mode

[SwLLR08] Regenerative-Braking Mode to EV Mode	
Requirement statement (shall)	When the active software mode is the regenerative-braking mode and P_dem >= PDEM_STOP_LOW && speed >

	SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_OUT, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is the regenerative-braking mode, after the higher-priority engine-supported and standstill exits have been excluded.
Derived from	SwHLR02, SwHLR05
Stakeholder Priority	High
Rationale	Implements the return from regenerative braking to EV propulsion.
Related UC(s)	UC2, UC4

4.3.9 SwLLR09 - Start Mode to Regenerative-Braking Mode

[SwLLR09] Start Mode to Regenerative-Braking Mode	
Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is Start mode. This is the highest-priority external exit from the engine-supported mode representation.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to RegenB exit to the Start-mode software branch.
Related UC(s)	UC4, UC5

4.3.10 SwLLR10 - ICE Mode to Regenerative-Braking Mode

[SwLLR10] ICE Mode to Regenerative-Braking Mode	
Requirement statement (shall)	When the active software mode is ICE mode and wEng > ENG_ON && speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is ICE mode. This is the highest-priority external exit from ICE mode.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to RegenB exit to the ICE-mode software branch.
Related UC(s)	UC4, UC5

4.3.11 SwLLR11 - Hybrid Mode to Regenerative-Braking Mode

[SwLLR11] Hybrid Mode to Regenerative-Braking Mode	
Requirement statement (shall)	When the active software mode is Hybrid mode and wEng > ENG_ON && speed > SPEED_REGEN && P_dem <=

	PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode. This is the highest-priority external exit from Hybrid mode.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to RegenB exit to the Hybrid-mode software branch.
Related UC(s)	UC4, UC5

4.3.12 SwLLR12 - Start Mode to EV Mode

[SwLLR12] Start Mode to EV Mode	
Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after the higher-priority regenerative-braking exit has been excluded. External exits are evaluated before internal Start-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to EV exit to the Start-mode software branch.
Related UC(s)	UC2, UC5

4.3.13 SwLLR13 - ICE Mode to EV Mode

[SwLLR13] ICE Mode to EV Mode	
Requirement statement (shall)	When the active software mode is ICE mode and wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after the higher-priority regenerative-braking exit has been excluded. External exits are evaluated before internal ICE-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to EV exit to the ICE-mode software branch.
Related UC(s)	UC2, UC5

4.3.14 SwLLR14 - Hybrid Mode to EV Mode

[SwLLR14] Hybrid Mode to EV Mode	
Requirement statement (shall)	When the active software mode is Hybrid mode and wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after the higher-priority regenerative-braking exit has been excluded. External exits are evaluated before internal Hybrid-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to EV exit to the Hybrid-mode software branch.
Related UC(s)	UC2, UC5

4.3.15 SwLLR15 - Start Mode to Standstill Mode

[SwLLR15] Start Mode to Standstill Mode	
Requirement statement (shall)	When the active software mode is Start mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after the higher-priority regenerative-braking and EV exits have been excluded. This is the third external exit checked before internal Start-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to StandStill exit to the Start-mode software branch.
Related UC(s)	UC5, UC6

4.3.16 SwLLR16 - ICE Mode to Standstill Mode

[SwLLR16] ICE Mode to Standstill Mode	
Requirement statement (shall)	When the active software mode is ICE mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after the higher-priority regenerative-braking and EV exits have been excluded. This is the third external exit checked before the remaining internal ICE-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High

Rationale	Allocates the chart-level Motion_mode_ICE to StandStill exit to the ICE-mode software branch.
Related UC(s)	UC5, UC6

4.3.17 SwLLR17 - Hybrid Mode to Standstill Mode

[SwLLR17] Hybrid Mode to Standstill Mode	
Requirement statement (shall)	When the active software mode is Hybrid mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after the higher-priority regenerative-braking and EV exits have been excluded. This is the third external exit checked before the remaining internal Hybrid-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to StandStill exit to the Hybrid-mode software branch.
Related UC(s)	UC5, UC6

4.3.18 SwLLR18 - Start Mode to Hybrid Mode

[SwLLR18] Start Mode to Hybrid Mode	
Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && SOC >= SOC_MID && (speed > SPEED_EV_MAX || P_dem >= PDEM_HYB_MID), the software shall transition to Hybrid mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after all higher-priority external exits have been excluded. This is the highest-priority remaining internal Start-mode transition.
Derived from	SwHLR02, SwHLR07
Stakeholder Priority	High
Rationale	Implements the internal Start_mode transition to Hybrid mode.
Related UC(s)	UC3, UC5

4.3.19 SwLLR19 - Start Mode to ICE Mode

[SwLLR19] Start Mode to ICE Mode	
Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && (SOC < SOC_MID || (speed <= SPEED_EV_MAX && P_dem < PDEM_HYB_MID)), the software shall transition to ICE mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after all higher-priority external exits and the higher-priority internal Hybrid-mode transition have been excluded.
Derived from	SwHLR02, SwHLR07
Stakeholder Priority	High

Rationale	Implements the internal Start_mode transition to ICE mode.
Related UC(s)	UC3, UC5

4.3.20 SwLLR20 - ICE Mode to Start Mode

[SwLLR20] ICE Mode to Start Mode	
Requirement statement (shall)	When the active software mode is ICE mode and wEng <= ENG_OFF, the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after all higher-priority external exits have been excluded. This is the highest-priority remaining internal ICE-mode transition.
Derived from	SwHLR02, SwHLR08
Stakeholder Priority	High
Rationale	Implements the engine-off return from ICE mode to Start mode.
Related UC(s)	UC3, UC5

4.3.21 SwLLR21 - ICE Mode to Hybrid Mode

[SwLLR21] ICE Mode to Hybrid Mode	
Requirement statement (shall)	When the active software mode is ICE mode and P_dem >= PDEM_HYB_MID && SOC >= SOC_MID, the software shall transition to Hybrid mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after all higher-priority external exits and the higher-priority internal Start-mode return have been excluded.
Derived from	SwHLR02, SwHLR08
Stakeholder Priority	High
Rationale	Implements the internal ICE_mode transition to Hybrid mode.
Related UC(s)	UC3, UC5

4.3.22 SwLLR22 - Hybrid Mode to Start Mode

[SwLLR22] Hybrid Mode to Start Mode	
Requirement statement (shall)	When the active software mode is Hybrid mode and wEng <= ENG_OFF, the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after all higher-priority external exits have been excluded. This is the highest-priority internal transition from Hybrid mode.
Derived from	SwHLR02, SwHLR09
Stakeholder Priority	High
Rationale	Implements the internal Hybrid_mode return to Start mode when engine speed falls to the engine-off threshold.
Related UC(s)	UC3, UC5

4.3.23 SwLLR23 - Hybrid Mode to ICE Mode

[SwLLR23] Hybrid Mode to ICE Mode

Requirement statement (shall)	When the active software mode is Hybrid mode and (P_dem <= PDEM_HYB_LOW || SOC < SOC_LOW), the software shall transition to ICE mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after all higher-priority external exits and the higher-priority internal Start-mode return have been excluded.
Derived from	SwHLR02, SwHLR09
Stakeholder Priority	High
Rationale	Implements the internal Hybrid_mode transition back to ICE mode.
Related UC(s)	UC3, UC5

5. NON-FUNCTIONAL REQUIREMENTS

This section defines non-functional constraints that apply to the same mode-logic baseline. These requirements complement Section 4 by constraining timing, stability, resource usage, and deterministic behavior of the implementation.

5.1 NfHLR01 - Periodic Execution and Sample Time

[NfHLR01] Periodic Execution and Sample Time	
Requirement statement (shall)	The software shall execute at a fixed periodic rate of 0.1 s consistent with the mode-logic sample-time calibration and resolve the next mode within the same sample period.
Conditions / Applicability	Applies to every periodic execution of the mode-logic software during nominal operation.
Stakeholder Priority	High
Rationale	Preserves the timing semantics of the source model and prevents backlog between control cycles.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

5.2 NfHLR02 - Transition Stability and Hysteresis Preservation

[NfHLR02] Transition Stability and Hysteresis Preservation	
Requirement statement (shall)	The software shall preserve the modeled hysteresis at the battery-state-of-charge boundary (entry at SOC_EV_IN, exit at SOC_EV_OUT) and at the engine-speed boundary (engine on at ENG_ON, engine off at ENG_OFF) while evaluating source-state exits in deterministic priority order.
Conditions / Applicability	Applies to all mode-transition evaluations, especially near calibrated boundaries.
Stakeholder Priority	High
Rationale	Prevents oscillation near thresholds and keeps the software behavior aligned with the Stateflow chart.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

5.3 NfHLR03 - Static Allocation and Bounded State Storage

[NfHLR03] Static Allocation and Bounded State Storage	
Requirement statement (shall)	The mode-logic implementation shall use only fixed-size pre-allocated storage, without dynamic memory allocation, recursion, or variable-length storage.
Conditions / Applicability	Applies to the mode-logic software components and the associated runtime data structures.
Stakeholder Priority	High
Rationale	Supports predictable resource usage and straightforward embedded deployment.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

5.4 NfHLR04 - Deterministic Outputs and Reproducibility

[NfHLR04] Deterministic Outputs and Reproducibility	
Requirement statement (shall)	Given the same initial mode and the same input sequence, the software shall produce the same sequence of operating modes and enable outputs across repeated executions of the same release baseline.
Conditions / Applicability	Applies to repeated runs of the same software baseline under the same nominal build configuration.
Stakeholder Priority	High
Rationale	Supports repeatable analysis, debugging, and regression comparison.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6